

Compile once. Read the current fact.

Ingest-Time Fact Compilation for Cost-Efficient and Reliable
Question Answering over Revised Corpora

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What does “it” mean when the passage is gone?

Question in the source: “Do you think that we are at the neutral rate?”

BEFORE / CHAIR POWELL

“I always had it, you know, between 3 and 4 percent.”



AFTER / ACTUAL GENERATED FACT

Chair Powell stated that his personal estimate of the neutral rate is between 3 and 4 percent.

“I” becomes a named speaker. “It” becomes a named concept.

A fact is a portable, source-grounded claim.

ONE CLAIM YOU CAN RETRIEVE ON ITS OWN

**Chair Powell stated that his personal estimate
of the neutral rate is between 3 and 4 percent.**

ATTRIBUTION

CONTENT

SOURCE

Powell's estimate

Neutral rate: 3–4%

FOMC · 29 Apr 2026

Another extracted claim: “the neutral rate cannot be known with certainty.”

Which launch decision is valid today?

Three versions of one project's launch-region decision.

Use the latest approval that is still in force.

REVISION 5

Approved.
Still in effect.

REVISION 11

Approved.
Then withdrawn.

REVISION 14

Draft.
Not approved.

Older versions → Newer versions

Which of these three can the answer rely on?

Revision 5 is still the valid decision.

The newer approval was withdrawn. The newest version is a draft.

REVISION 5 / USE THIS

Approved.
Still in effect.

REVISION 11 / EXCLUDE

Approval
withdrawn.

REVISION 14 / EXCLUDE

Draft.
Not approved.

KEEP THE SOURCE WITH THE FACT

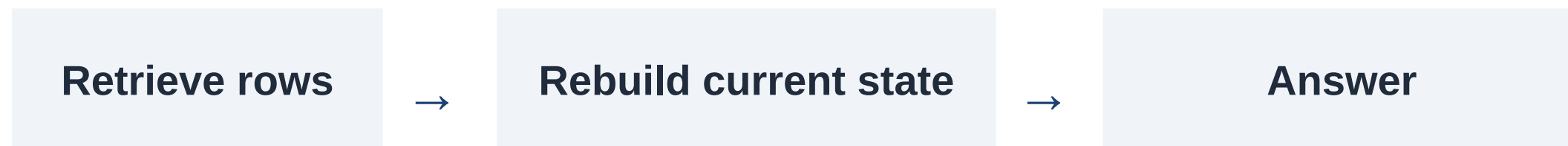
Launch-region decision

Revision 5 · owner-approved-r5.md

Resolve the rule once. Reuse the current fact.

Every new question should reuse that work.

RECONSTRUCT AT READ TIME



...again for the next question.

COMPILE AT INGEST / CHANGE TIME



We changed the reader's job, not its model.

RAW-CONTEXT INPUT / QSR

64 noisy rows
+ governance rules

RESOLVED INPUT / ISC

1 current row
+ the source behind it

Same inexpensive model. Same requested answer format.

Success means all three match:

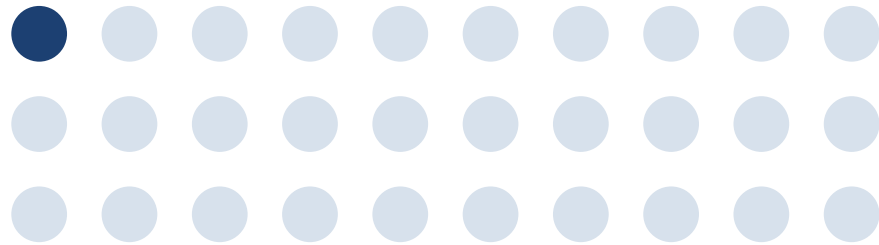
the value, the source, and the revision.

Control: the resolved row is supplied from known answers; compiler accuracy is not measured.

Same model: 1/30 exact answers → 30/30.

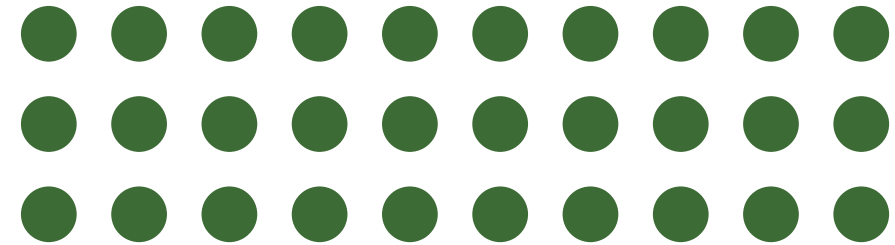
RECONSTRUCT AT QUERY TIME

1/30



READ THE RESOLVED ROW

30/30



29 QSR runs exhausted the completion budget
before producing a parseable final answer.

When both were right, one read 21.6× fewer tokens.

Same correct answers on the simpler revision questions.



Less to reconstruct.

14.21×

Less to pay for on every read.

lower mean read cost

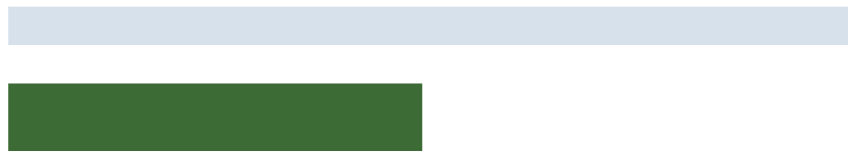
Not every corpus is equal.

HUMAN DIALOGUE

Federal Reserve Q&A · 25 passages

49%

of original tokens after extraction



Speech leaves repetition,
fillers and implied context.

WIKIPEDIA CONTENT

Encyclopedic articles · 30 passages

103%

of original tokens after extraction



Already “compiled” by human editors:
less fluff, clearer language.

Gray: source text. Color: extracted facts. Editorial explanation is an interpretation.

Shorter can still leave out the point.

SOURCE / NICK TIMIRAOS

“The statement today preserves language that has taken on some meaning, as it was socialized when the Committee was actively lowering rates.”

2 OF 6 GENERATED FACTS

Core inflation is at 3.2 percent.

The Committee needs to see more data.

But did the statement's language actually change?

The full six-fact output lost the unchanged-language claim.

Make reusable interpretation a durable asset.

Resolve the reference.

Resolve the revision.

Then let each question read the fact
and inspect where it came from.

Next: prove the full loop on real, continuously revised sources.

Code + frozen artifacts github.com/aix-sc/isc

Appendix: the two recorded responses

QSR / RAW COMPLETION EXCERPT

First, filter rows that match
entity='adversarial_project_000' and
slot='launch_region'.

[continues listing candidates]

No parseable final answer.
1,024 completion tokens.

ISC / ANSWER FIELDS

value: launch_region_governed_000
revision: 5
source: .../owner-approved-r5.md

Value + source + revision exact.
221 completion tokens.

Trace is excerpted; source path abbreviated. Full inputs and outputs: examples.json.

Appendix: measurements behind the headline

ADVERSARIAL / FLASH

QSR 1/30 → ISC 30/30 exact

Cost/q \$0.00131 → \$0.00010 (12.89×)

Latency 8.87 s → 3.06 s

SIMPLE REVISIONS / FLASH

Both exact; tokens/q 9,777 → 453 (21.6×)

Cost/q \$0.00142 → \$0.00010 (14.21×)

Latency 3.40 s → 3.30 s

S2 / DIALOGUE

25 passages; token ratio 0.49×

Entailed 97.6%; QA fidelity 92.0%

Cost/fact \$0.00003 (reconstructed)

S2 / PROSE

30 passages; token ratio 1.03×

Entailed 100.0%; QA fidelity 96.7%

Cost/fact \$0.00001 (reconstructed)

Appendix: what remains to be tested

THE READ-PATH CONTROL

Reference-resolved rows are supplied from expected fields.

The end-to-end compiler is not tested.

THE MISSING COMPARISON

A deterministic row filter.

More time to finish an answer.

Real revision streams.

HEAVIER-MODEL SPOT CHECKS / THREE ADVERSARIAL QUESTIONS × FIVE SEEDS

Kimi K2.7 Code QSR: 6.7% mean exact; \$0.01086/question.

GLM 5.2 QSR: 73.3% mean exact; \$0.01389/question.

Pro simple-revision sample: all exact; \$0.01851/question (185.10× Flash ISC).

Appendix: where each example comes from

Paper: Wild, Takahashi & Uraki, iCAST-ES 2026.

Camera-ready: aix-sc/isc @ bd7b8c5, paper/icast2026-camera-ready/main.tex.

Real examples: Federal Reserve press conference, 29 Apr 2026.

Frozen S2 passage 17: Powell's neutral-rate estimate.

Frozen S2 passage 4: omitted unchanged-language claim.

Synthetic example: adversarial_project_000:launch_region, seed 20260709.

Run audit: CAMERA_READY_EVIDENCE_2026-08-20.md.

Template: committee-supplied "Template iCAST 2026.pptx".
